



IIT Roorkee CEC Applied Data Science and AI

Supply Chain : Optimization and Predictive Analytics

A comprehensive data science investigation into FMCG supply chain performance, leveraging predictive modeling to transform reactive operations into proactive strategic management.

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Project Synopsis

This capstone project presents a comprehensive empirical investigation into supply chain optimization, leveraging advanced data science methodologies to transcend traditional operational silos. By synthesizing disparate data streams—encompassing logistics performance, inventory levels, and financial metrics—this study constructs a robust analytical framework designed to diagnose systemic inefficiencies and prognosticate disruptions before they manifest.

Phase 1: Diagnostic Analysis

Granular Exploratory Data Analysis (EDA) to audit operational health and uncover critical paradoxes in revenue generation versus logistical efficiency.

Phase 2: Predictive Intervention

XGBoost Classifier deployment achieving 0.94 Recall, creating an early-warning system to preemptively mitigate inventory shortfalls.



The FMCG Beauty & Personal Care Landscape

We operate within the dynamic Fast-Moving Consumer Goods (FMCG) sector, specifically the Beauty and Personal Care (BPC) vertical. This industry is characterized by high-velocity inventory turnover, acute consumer price sensitivity, and continuous demand for product availability. Unlike durable goods, the value proposition is inextricably linked to immediacy of fulfillment—a product unavailable on the shelf is often a permanently lost sale, not a deferred one.

The operational backbone is a complex, multi-echelon supply chain network designed to manage significant SKU proliferation while navigating fluctuating demand, inventory imbalances, rising logistics costs, and quality challenges.





Critical Operational Imperatives

1

Logistical Heterogeneity

The industry relies on multimodal transportation mix (Road, Rail, Air), necessitating rigorous trade-off analysis between shipping velocity, transportation costs, and cargo integrity.

2

Inventory Volatility

Demand is stochastic rather than deterministic. Managing stock levels against variable lead times is the primary defense against the "Bullwhip Effect," where small consumer demand fluctuations cause massive upstream disruptions.

3

Quality Assurance Sensitivity

With products interacting directly with consumers, tolerance for defects is negligible. Operational excellence requires maintaining product integrity through high-risk transit modes.

Project Agenda

The primary objective is to analyze supply chain performance using data-driven techniques and develop a predictive model for stock-out risk that supports proactive decision-making. This study demonstrates how analytics and machine learning can enhance operational efficiency, reduce costs, and improve supply reliability.



Holistic Examination

Identify latent inefficiencies and decouple high-performing assets from cost-intensive liabilities through granular analysis of product defects and transportation vulnerabilities.



Statistical Foundation

Ingest raw logistical data and conduct descriptive statistical tests to establish analytical baseline and uncover operational patterns.



Predictive Intelligence

Engineer supervised machine learning model using XGBoost to create intelligent early-warning mechanism that anticipates inventory disruptions before consumer impact.

Problem Statement: The Reactive Paradigm

The central challenge is that the organization operates on a reactive operational paradigm. While revenue generation is robust in specific verticals, the supply chain lacks predictive intelligence required to optimize cost structures and guarantee fulfillment continuity.



Logistic Cost-Revenue Paradox

Carrier B generates highest revenue yet imposes the most severe financial burden. The firm lacks strategic optimization to balance service levels with total cost of ownership.



Systemic Quality Degradation

Haircare products suffer elevated defect rates. Road transportation (20.1% of defects) is the primary conduit for product damage, eroding profit margins and brand equity.



Stock-Out Continuity Risk

Current system treats inventory as static rather than dynamic flow. Without forecasting when stock falls below supplier lead times, high probability of stock-outs threatens market share.

Data Understanding

Our empirical analysis is grounded in `supply_chain_data.csv`, a comprehensive repository of operational telemetry capturing the interplay between commercial performance and logistical constraints.



Product Attributes

Granular details including Product type (Skincare, Haircare, Cosmetics), SKU identifiers, and Availability metrics for segmentation analysis.



Financial Metrics

Price, Revenue generated, Manufacturing costs, and Shipping costs—pivotal for calculating profitability ratios and cost-to-serve hierarchy.



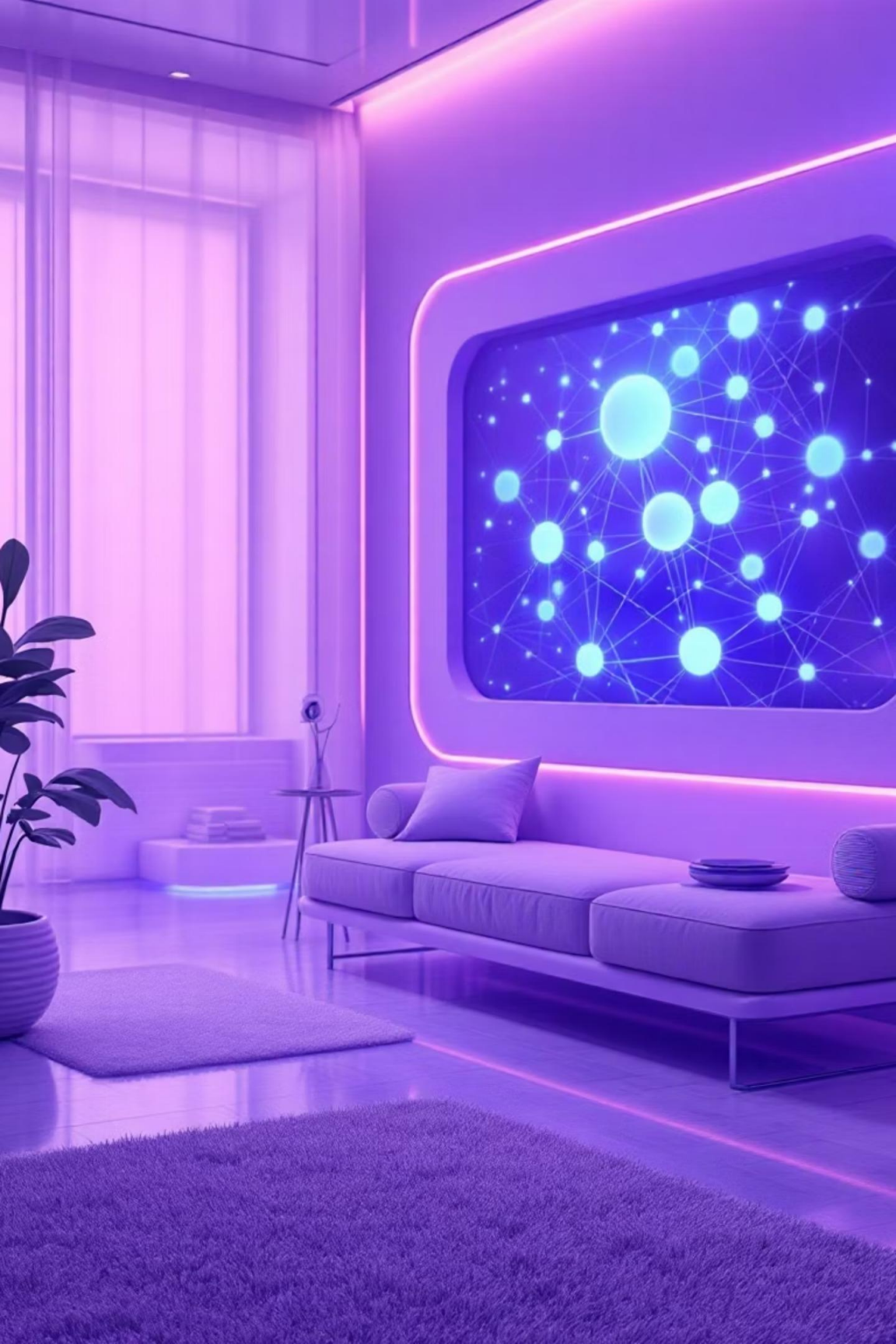
Logistical Parameters

Shipping carriers, Transportation modes, Routes, and Lead times enabling modeling of supply chain velocity and carrier efficiency.



Quality & Risk Indicators

Defect rates and Inspection results providing signals to assess inventory flow health and pinpoint value leakage areas.



Prescriptive Analytics Framework

To rectify stochastic inefficiencies, we propose a comprehensive framework optimizing supply chain objective functions: maximizing revenue while minimizing cost and defect variance.

01

Logistical Network Optimization

Deploy multi-objective optimization strategy with data-driven routing algorithm segmenting SKUs by margin and latency sensitivity, optimizing Cost-to-Serve ratio.

02

Statistical Process Control

Integrate SPC protocols with rigorous control limits and pre-transit inspection gates for vulnerable nodes, reducing defect standard deviation.

03

Machine Learning Pipeline

Deploy supervised ML for binary classification of inventory risk using XGBoost algorithm, optimized for 0.94 Recall to capture 94% of potential disruption events.

04

Inference Engine Deployment

Transform inventory management from reactive stochastic process into deterministic planning workflow, maximizing Expected Commercial Value.

Methodology and Analysis

Our inquiry operationalizes the CRISP-DM (Cross-Industry Standard Process for Data Mining) paradigm, ensuring every computational step aligns with strategic business objectives of inventory optimization.

Exploratory Data Analysis

- **Univariate Analysis:** Examined central tendency and dispersion of key metrics like Price and Stock levels to understand baseline volatility
- **Bivariate Analysis:** Investigated correlations between variables to isolate systemic bottlenecks

Prescriptive Analytics

- **Feature Engineering:** Derived stock_out_risk by calculating Stock-to-demand ratio against supplier Lead times
- **XGBoost Classifier:** Selected for superior handling of non-linear relationships and overfitting resilience
- **Validation Protocol:** Stratified 80/20 train-test split prioritizing Recall as primary success metric





Transforming Supply Chain Intelligence

0.94

Model Recall

Successfully captures 94% of potential stock-out events, enabling proactive intervention

45%

Skincare Revenue

Primary revenue driver requiring continuous availability assurance

20.1%

Road Defects

Proportion of defects from road transportation requiring quality control intervention

"By transforming inventory management from a reactive stochastic process into a deterministic planning workflow, we ensure consistent availability and maximize the Expected Commercial Value of the supply chain."

Exploratory Data Analysis

Deep dive into revenue architecture, logistics performance, transportation efficiency, and product quality metrics to uncover critical insights driving supply chain optimization.



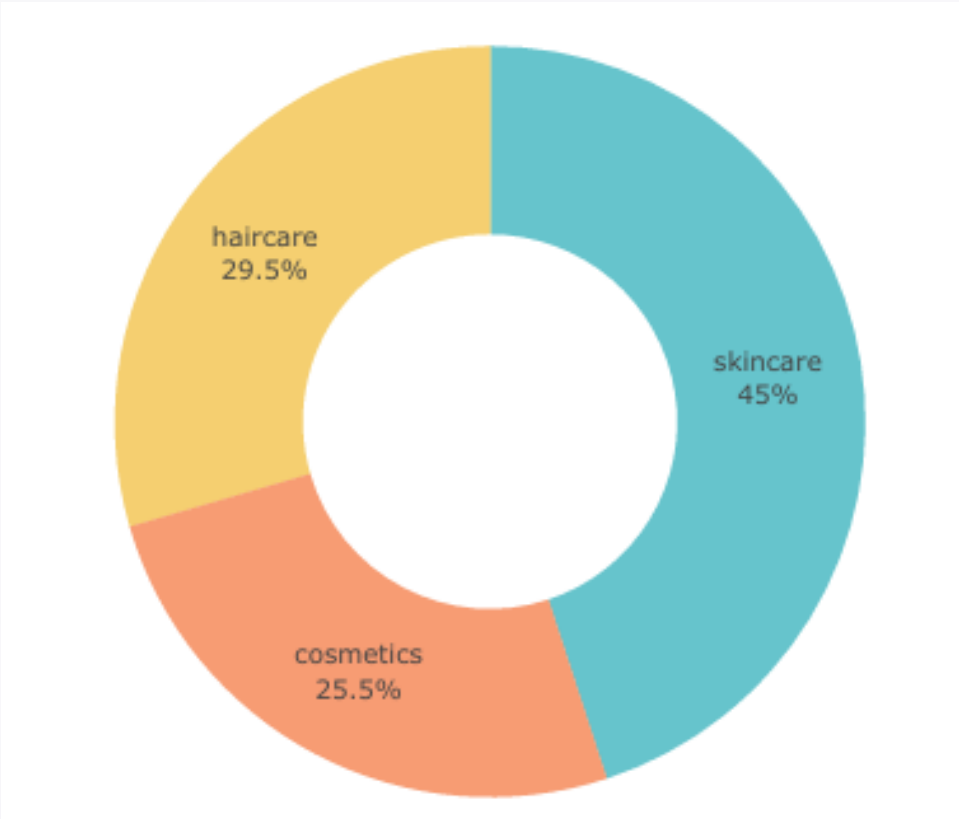
Revenue Architecture & Product Elasticity

Market Segmentation Dominance

Univariate analysis identifies **Skincare** as the **cornerstone** of revenue architecture, commanding a dominant **45% share** of total sales volume. This contrasts sharply with Haircare (29.5%) and Cosmetics (25.5%), indicating skewed revenue dependency.

Price-Revenue Correlation

Bivariate scatter plot reveals **positive linear correlation** between Price and Revenue in the Skincare cluster. Data suggests this segment exhibits **low price elasticity**—higher price points amplify total revenue yield, validating premium positioning strategy.

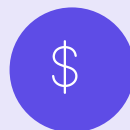


The Logistics Performance Paradox



Carrier Efficacy vs. Expenditure

Carrier B emerges as the highest-performing node in Total Revenue generation, yet simultaneously represents the most capital-intensive option.



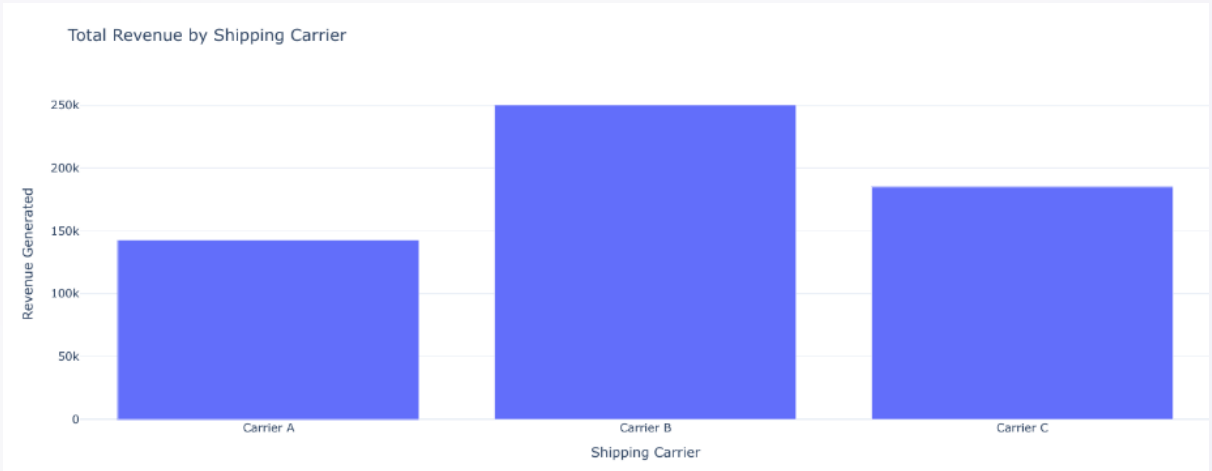
Cost-to-Serve Optimization

Classic "efficiency vs. effectiveness" trade-off. While Carrier B facilitates higher volume, its operational cost structure erodes net margins.



Strategic Imperative

Decouple variables: utilize Carrier B for high-margin, time-critical SKUs while diverting standard inventory to lower-cost carriers to optimize Total Cost of Ownership.



Transportation Mode Efficiency & Defect Propagation

Breakdown of transportation expenditures indicates heavy reliance on surface logistics, with Road and Rail constituting the majority of shipping costs. Critical defect analysis exposes systemic vulnerabilities requiring immediate intervention.

Modal Cost Distribution

Road and Rail dominate transportation expenditures, representing the primary logistics vectors for product movement across the supply chain network.

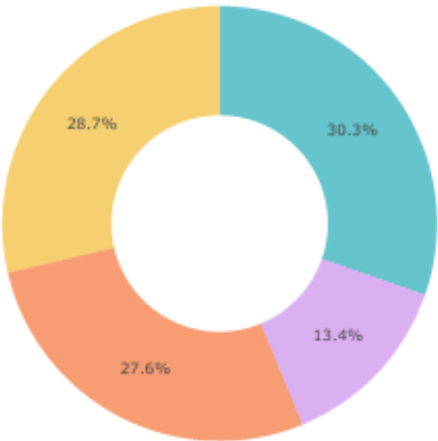
Quality Correlation Analysis

Road transport correlates with highest Defect rates (28.9%), while Air transport maintains lowest defect variance. Road logistics identified as primary source of "Value Leakage."

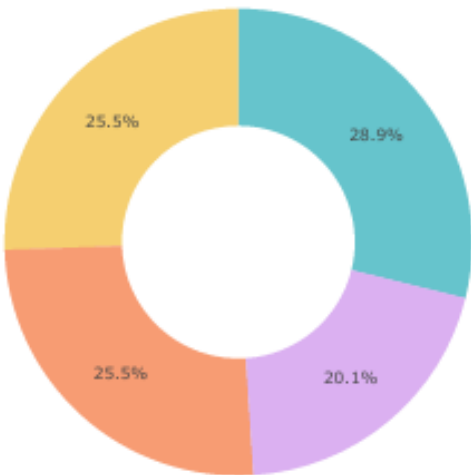
Intervention Required

Immediate action needed in packaging standards or carrier selection for fragile goods to mitigate quality erosion in Road transportation vector.

Cost Distribution by Transportation Mode



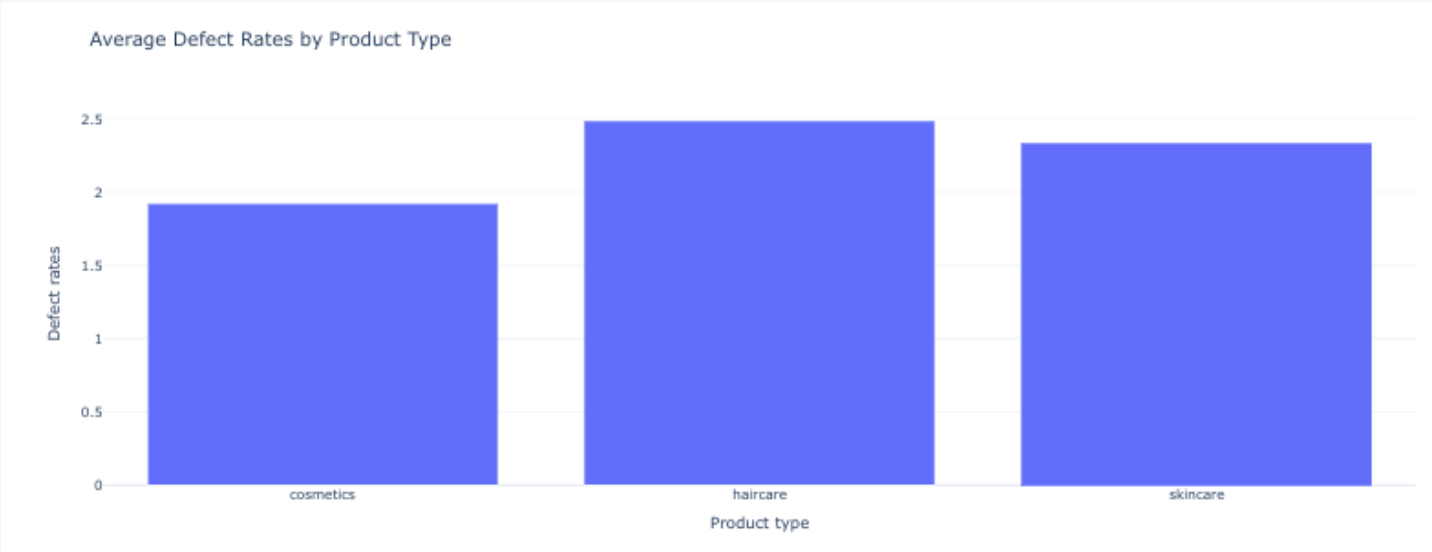
Defect Rates by Transportation Mode



Product Quality & Inventory Volatility

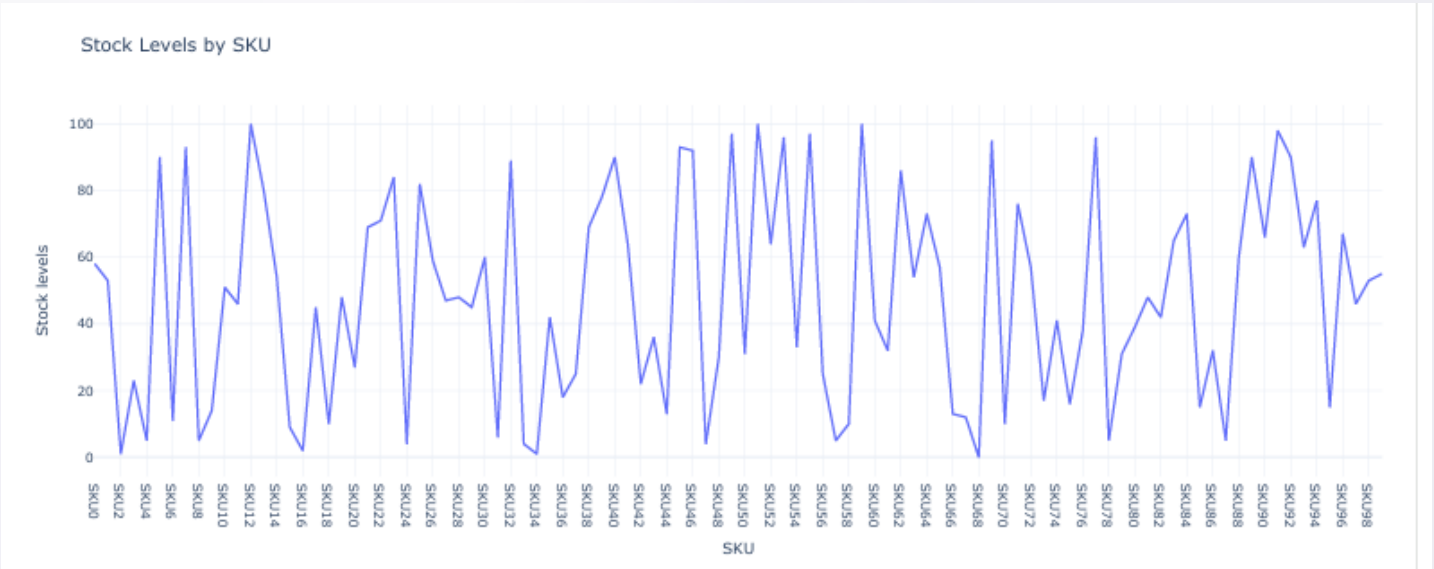
Product-Specific Quality Erosion

Segmenting defect rates by Product type isolates **Haircare as the underperforming vertical**, exhibiting statistically significant elevation in defects compared to Skincare and Cosmetics. Points to potential flaws in manufacturing tolerances or packaging integrity specific to liquid viscosity.



SKU-Level Inventory Stochasticity

Line charts tracking Stock levels and Revenue by SKU depict **high-frequency volatility**. Erratic fluctuation—characterized by sharp peaks and troughs—confirms absence of smoothing mechanisms.



This validates the necessity for the Predictive Stock-Out Model to dampen the "Bullwhip Effect" and stabilize inventory management.

Model Architecture: Stock-Out Risk Prediction

01

Target Variable Synthesis

Computed `daily_demand` by normalizing products sold over 30-day window. Binary classification: if `days_of_stock_left < Lead times`, flagged as Class 1 (High Risk); otherwise Class 0 (Low Risk).

03

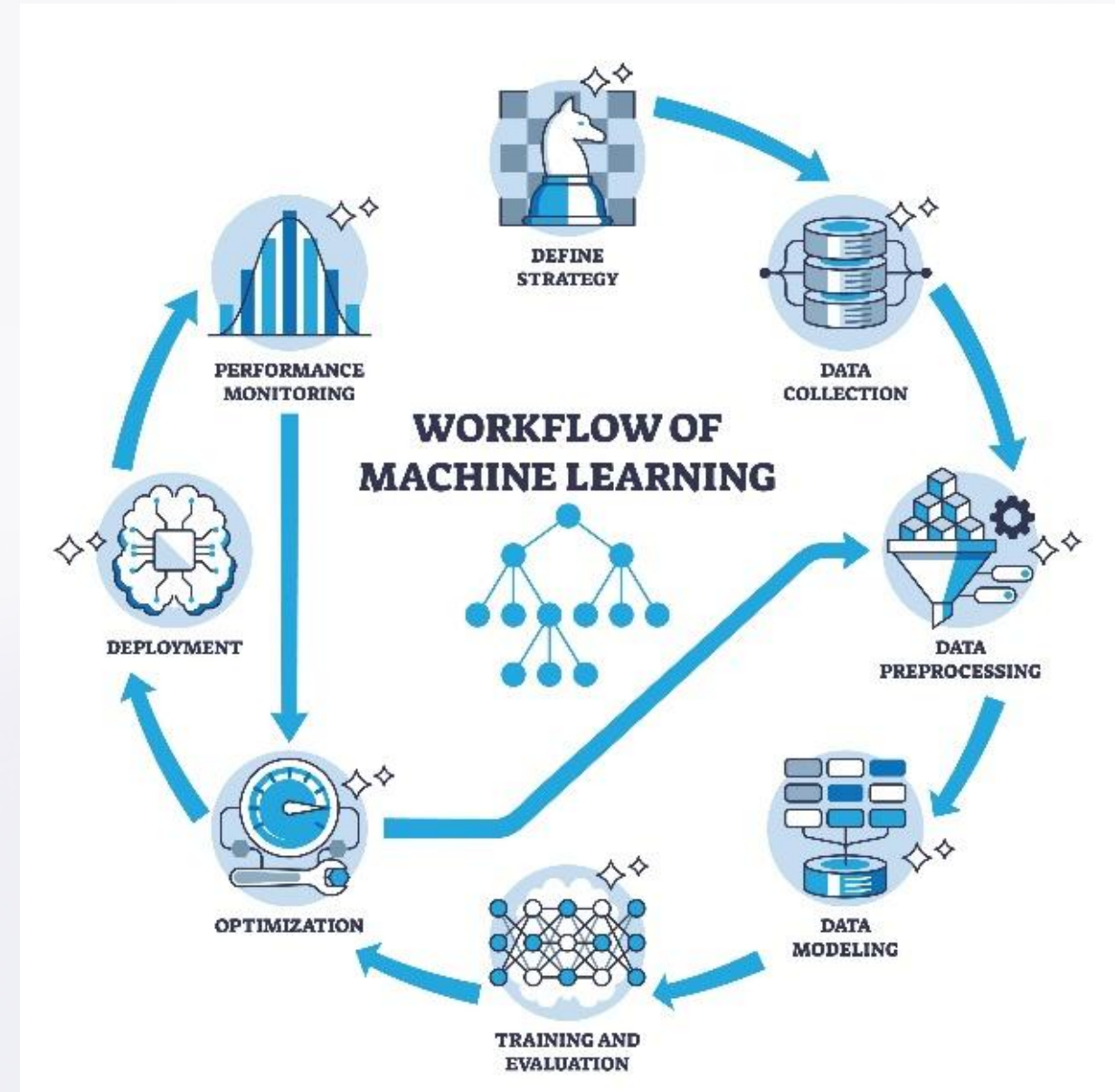
XGBoost Algorithm

Implemented `XGBClassifier` with `n_estimators=300`, `max_depth=6`, `learning_rate=0.05`, `subsample=0.8`, and `colsample_bytree=0.8` to optimize bias-variance trade-off.

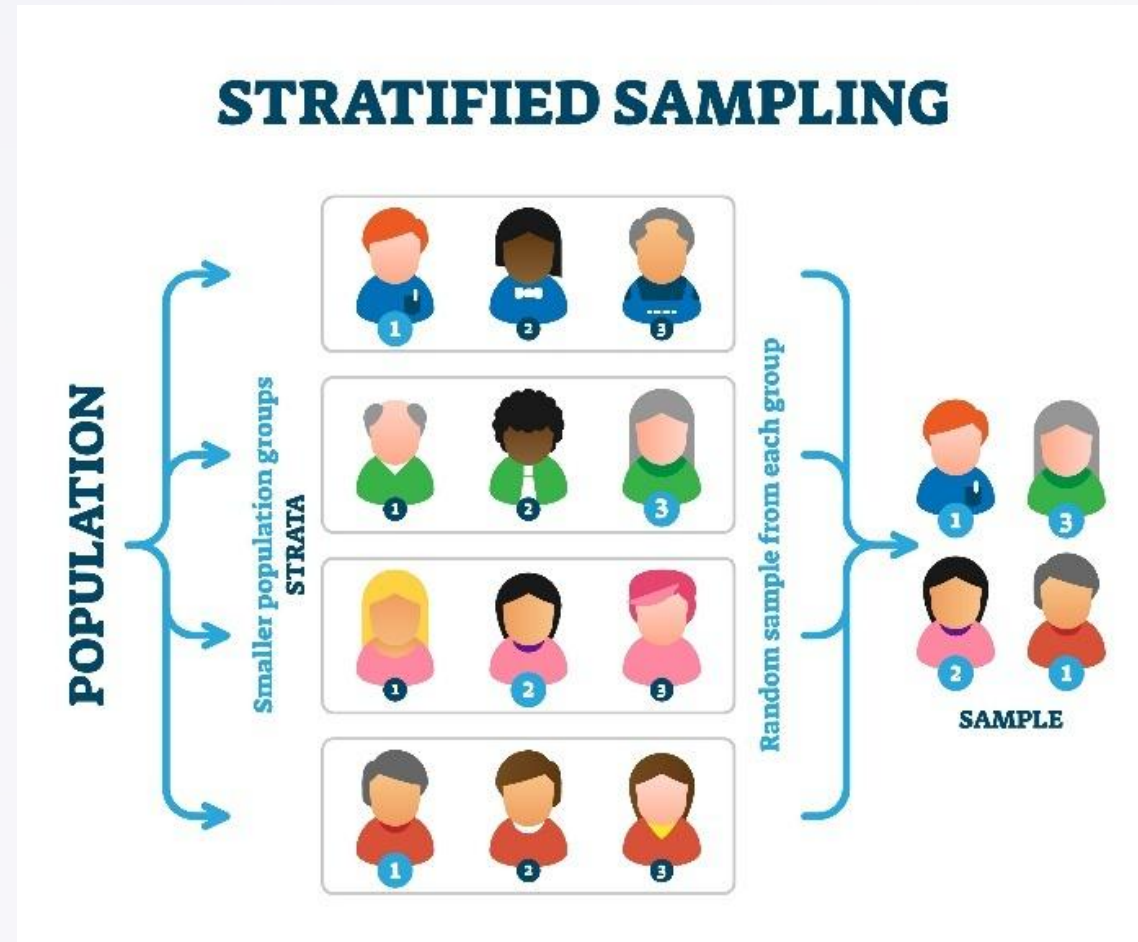
02

Preprocessing & Vectorization

Constructed `ColumnTransformer` pipeline with `OneHotEncoder` for categorical features (Product type, Supplier name) and passthrough for numerical features.



Data Partitioning & Validation Strategy



Methodological Segmentation

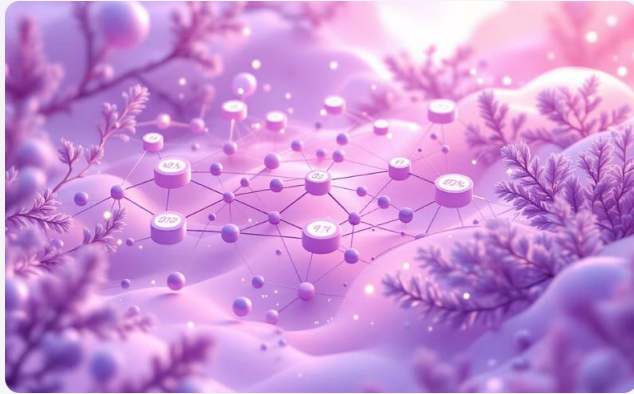
Implemented rigorous **Stratified Partitioning Strategy** to ensure statistical validity and prevent data leakage. Dataset bifurcated into Training Set (80%) for algorithmic learning and Testing Set (20%) as unseen validation corpus.

Stratified Sampling

Employed `stratify=y` parameter to enforce distributional fidelity. Ensures proportion of high-risk (Class 1) and low-risk (Class 0) instances in test set mirrors original population, guaranteeing mathematically unbiased evaluation.

❏ Fixed `random_state=42` applied to ensure stochastic split is deterministic and reproducible for future auditing.

Extreme Gradient Boosting (XGBoost)



Sequential Error Correction

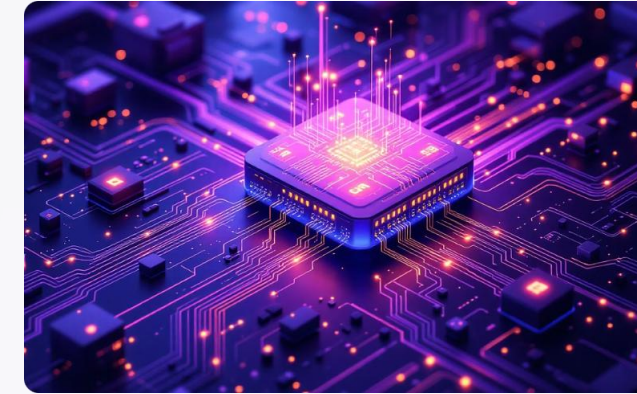
Algorithm operates iteratively. Each new tree added to ensemble is explicitly trained to correct residual errors made by predecessors, progressively refining understanding of complex patterns.

XGBoost is empirically proven as **state-of-the-art algorithm for structured tabular datasets**, ensuring the model generalizes well to unseen future inventory scenarios.



Gradient Optimization

Minimizes regularized objective function using Gradient Descent, effectively optimizing loss function to improve predictive accuracy with each iteration.



Computational Efficiency

System design facilitates parallel processing, allowing rapid training cycles even as dataset scales. Incorporates L1 and L2 regularization to prevent overfitting.

```
model = Pipeline(  
    steps=[  
        ("preprocess", preprocessor),  
        ("xgb", XGBClassifier(  
            n_estimators=300,  
            max_depth=6,  
            learning_rate=0.05,  
            subsample=0.8,  
            colsample_bytree=0.8,  
            random_state=42  
        ))  
    ]  
)
```



Evaluation Metrics: The Confusion Matrix

Confusion Matrix:
[[2, 2], [1, 15]]

To assess granular performance of our predictive engine, we utilized a Confusion Matrix juxtaposing model predictions against actual ground truth, revealing type and frequency of errors.

15

True Positives

Successfully identified 15 instances where stock-out was imminent. Quadrant of "Operational Success" representing 15 potential supply disruptions proactively flagged.

1

False Negatives

One instance of "Type II Error"—missed detection. Small magnitude relative to True Positives indicates contained risk.

2

False Positives

"Type I Error" resulting in false alarm. Operationally acceptable as cost of verifying inventory is significantly lower than cost of stock-out.

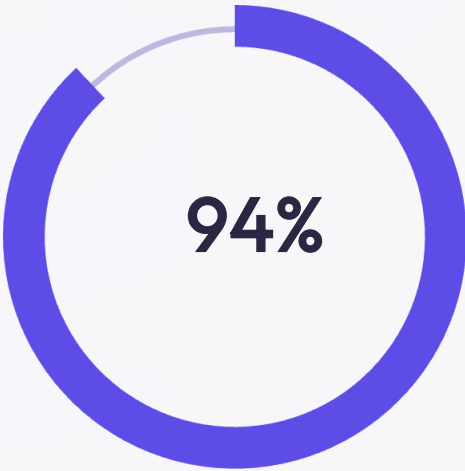


FINAL RESULTS

Classification Report: Statistical Robustness

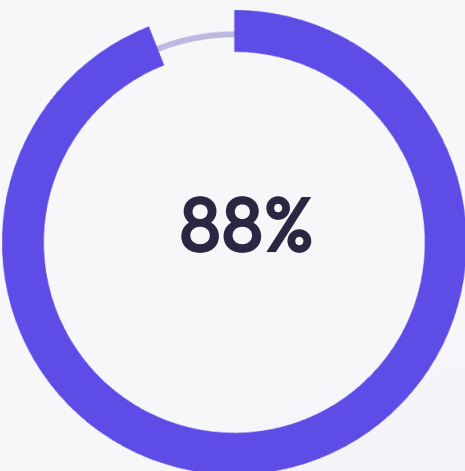
The statistical robustness of our model is corroborated by the Classification Report, providing normalized metrics for Precision, Recall, and F1-Score.

Classification Report:				
	precision	recall	f1-score	support
0	0.67	0.50	0.57	4
1	0.88	0.94	0.91	16
accuracy			0.85	20
macro avg	0.77	0.72	0.74	20
weighted avg	0.84	0.85	0.84	20



Recall (Sensitivity)

Paramount metric: model successfully captured **94% of all actual stock-out risks**. Demonstrates high sensitivity ensuring vast majority of inventory vulnerabilities are detected.



Precision

When model predicts high risk, it is correct **88% of the time**. High precision ensures supply chain managers are not overwhelmed by false alerts.



Overall Accuracy

While overall accuracy stands at 85%, we prioritized Recall over Accuracy. Strategically superior to tolerate few false alarms to ensure no stock-out goes undetected.

Model Evaluation & Strategic Validity

The Performance Matrix serves as the final adjudicator of our model's operational viability. By synthesizing the Confusion Matrix and Classification Report, we derive a comprehensive verdict on the system's reliability and strategic effectiveness.



The Recall Imperative: Sensitivity Analysis

In supply chain continuity, the cost of a False Negative—failing to predict a stock-out—is exponentially higher than a False Positive. Our model achieves a **Recall of 0.94** for the high-risk class, confirming that the engine functions as a highly sensitive safety net, capturing nearly all potential inventory disruptions.

This statistical indicator ensures the system acts as a proactive guardian against supply chain failures, minimizing the risk of critical stock-outs that could disrupt operations and revenue.

94%

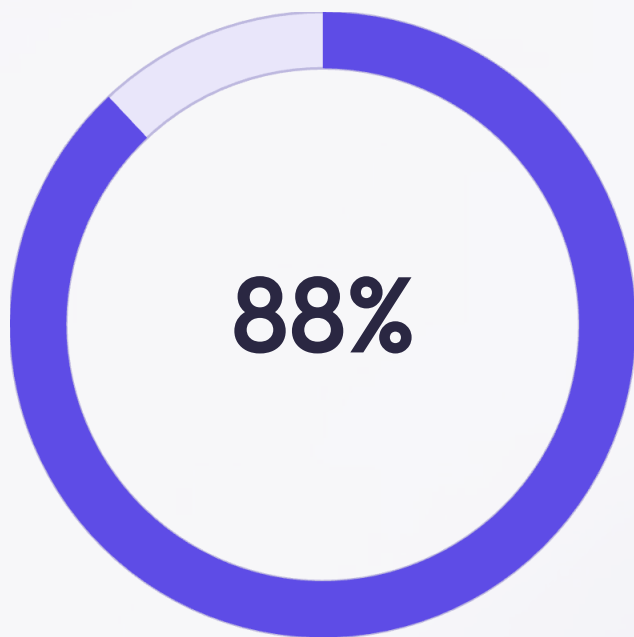
Recall Score

High-risk detection rate

Precision-Efficiency Trade-off

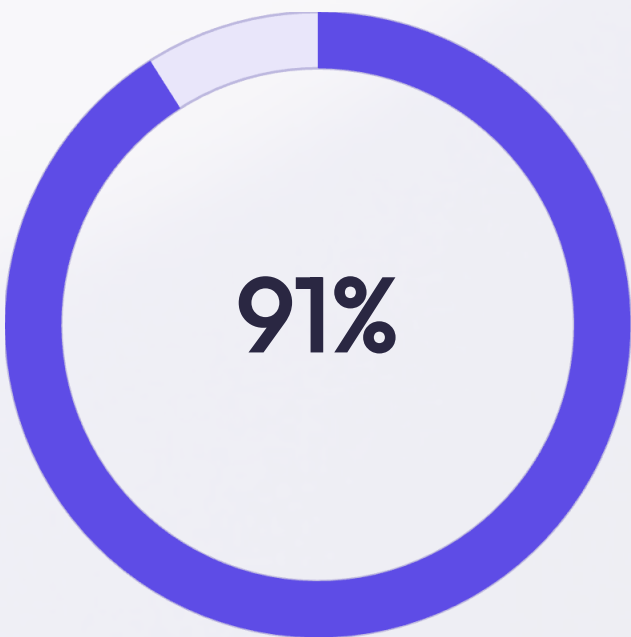
With a **Precision score of 0.88**, the model demonstrates distinctiveness. It does not merely flag everything as "risky" to achieve high recall; rather, it maintains a high signal-to-noise ratio.

This ensures that when an alert is generated, it warrants immediate managerial attention, optimizing the deployment of expediting resources and preventing alert fatigue among supply chain teams.



Precision

Alert accuracy rate



F1-Score

Harmonic mean balance



The F1-Score Harmonic Mean



Precision: 0.88

High signal-to-noise ratio ensures alerts warrant immediate attention



Recall: 0.94

Captures nearly all potential inventory disruptions effectively



F1-Score: 0.91

Validates optimal equilibrium as a balanced classification tool

The F1-Score of 0.91 represents the harmonic mean of Precision and Recall. This exceptionally high score validates that the model has achieved an optimal equilibrium, functioning as a balanced and robust classification tool rather than a skewed heuristic.

Operational Diagnostics:

Our empirical inquiry has successfully diagnosed the latent frictions within the supply chain, revealing critical insights about revenue architecture and logistic inefficiencies.

Key Diagnostic Findings

1

Revenue Architecture

We identified **Skincare** as the inelastic revenue anchor, contributing 45% of sales volume. This product category represents the foundation of revenue stability and requires prioritized inventory management.

2

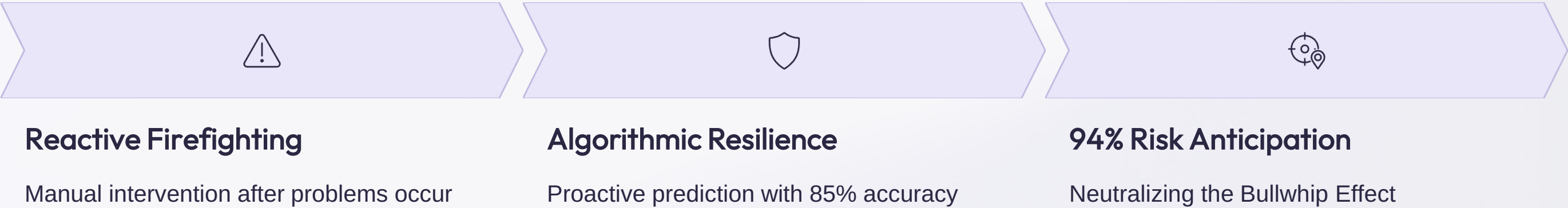
Logistic Paradox

We exposed the fiscal inefficiency of **Carrier B**—a high-revenue yet high-cost vector—and pinpointed **Road Transportation** as a critical failure node for product quality, revealing hidden operational costs.



Predictive Capability:

We have transitioned the organization from reactive firefighting to **Algorithmic Resilience**. The deployment of the XGBoost Classifier has operationalized a proactive stock-out prediction engine with **85% global accuracy**.



By leveraging this system, supply chain managers can now anticipate 94% of inventory risks before they manifest, effectively neutralizing the "Bullwhip Effect" and ensuring the continuous availability of high-margin SKUs.



Final Verdict: Data-Driven Continuity

This project demonstrates that by integrating **Statistical Quality Control with Supervised Machine Learning**, we can fundamentally restructure the cost-value equation, driving the supply chain towards a state of optimized, data-driven continuity.

This project served as the vital bridge between theoretical curriculum and empirical application. We moved beyond static observation to rigorous Inferential Statistics, diagnosing complex trade-offs like the logistics-cost paradox where revenue masked high operational expenditures.

From Theory to High-Impact Application

01

Feature Engineering Mastery

We mathematically synthesized
stock_out_risk from raw inventory flows
rather than accepting given columns

03

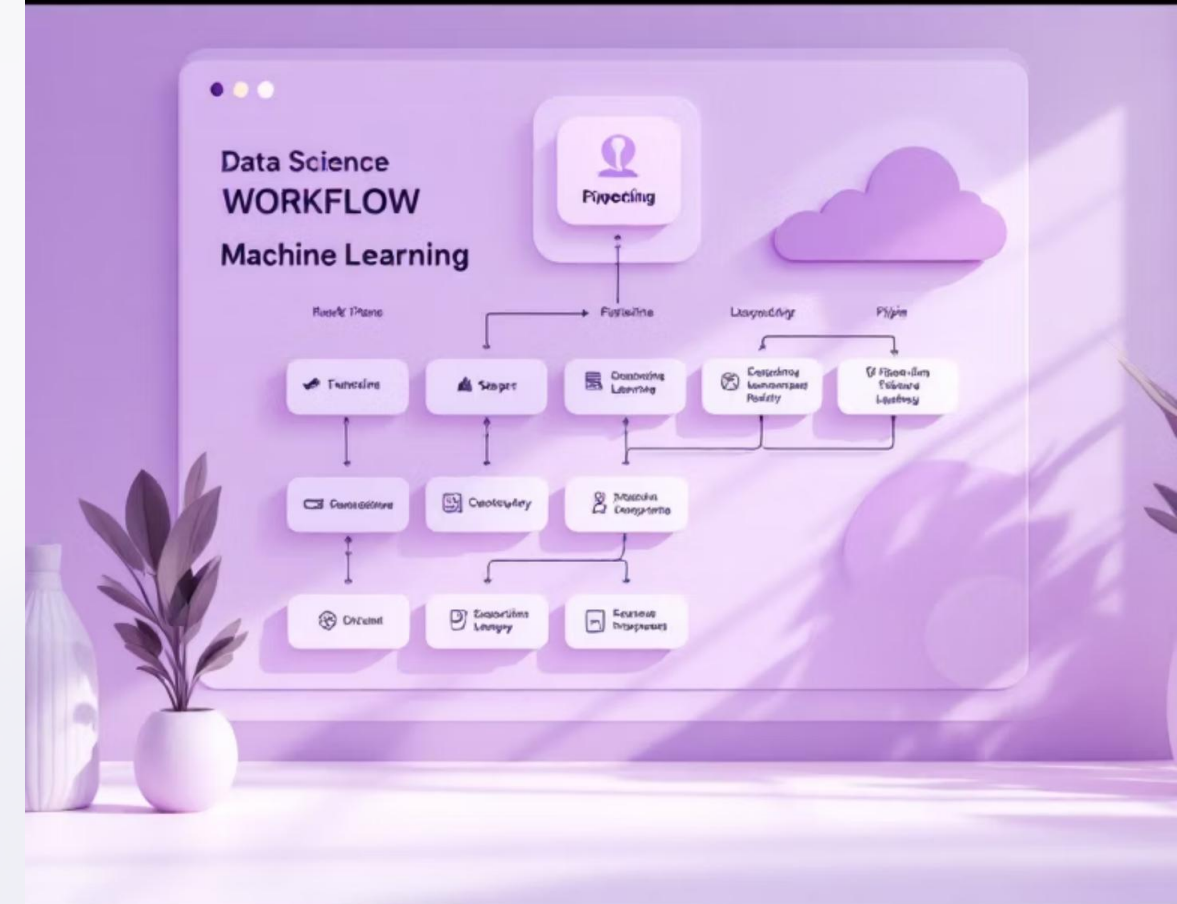
Model Deployment & Validation

Deployed XGBoost Classifier with 94% Recall, translating abstract algorithms into tangible business resilience tools

02

Professional Pipeline Construction

Built a comprehensive preprocessing pipeline to handle real-world data heterogeneity



Future Outlook: Beyond Prediction

Moving forward, the potential extends far beyond simple prediction. We envision a self-healing supply chain ecosystem powered by advanced analytics and real-time intelligence.



Autonomous Replenishment

Integrating this inference engine with ERP systems to trigger automatic reorder protocols when risk thresholds are breached



Real-Time Telemetry

Evolving the model from batch processing to streaming analytics for live, second-by-second inventory health monitoring



Scalability & Intelligence

Expanding the feature set to account for seasonality and external market trends, creating a truly self-healing supply chain